



Post Office Box 2000, Decatur, Alabama 35609-2000

September 9, 2025

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 2
Renewed Facility Operating License No. DPR-52
NRC Docket No. 50-260

Subject: **Licensee Event Report 50-260/2025-001-01**

Reference: **Letter from TVA to NRC, "License Event Report 50-260/2025-001-00," dated July 10, 2025 (ML25191A210)**

The enclosed Licensee Event Report provides details of a SCRAM due to low electro-hydraulic control pressure on Browns Ferry Nuclear Plant, Unit 2, followed by a second automatic Reactor Protection System actuation due to a low reactor water level transient caused by manually opening and closing a main steam relief valve. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as any event or condition that resulted in a manual or automatic actuation of the Reactor Protection System and Primary Containment Isolation System. Revision 1 to this report includes information from the cause evaluation that was not previously reported.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David J. Renn, Site Compliance Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "DLAK", is written over a horizontal line.

Daniel A. Komm
Site Vice President

Enclosure: Licensee Event Report 50-260/2025-001-01 – Reactor SCRAM due to Low Electro-Hydraulic Control Pressure

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cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Unit 2**

Licensee Event Report 50-260/2025-001-01

Reactor SCRAM due to Low Electro-Hydraulic Control Pressure

See Enclosed

NRC FORM 366 (04-02-2024)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 04/30/2027			
		LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)				Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov , and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk ail: pira_submission@omb.eop.gov . The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.			
1. Facility Name Browns Ferry Nuclear Plant, Unit 2				☒ 050 ☐ 052		52. Docket Number 00260			
						3. Page 1 OF 7			
4. Title Reactor SCRAM due to Low Electro-Hydraulic Control Pressure									
5. Event Date			6. LER Number			7. Report Date			
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	
05	12	2025	2025	- 001 -	01	09	09	2025	
						8. Other Facilities Involved			
						Facility Name	Docket Number		
						N/A	05000XXX		
						Facility Name	Docket Number		
						N/A	05000XXX		
9. Operating Mode 1				10. Power Level 100					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)									
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)		10 CFR Part 21		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☐ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)				☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)			
☐ OTHER (Specify here, in abstract, or NRC 366A).									
12. Licensee Contact for this LER									
Licensee Contact Justin Garner, Site Compliance Engineer								Phone Number (Include area code) 256-729-7955	
13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4. Supplemental Report Expected)					15. Expected Submission Date				
☒ No									
☐ Yes (If yes, complete 15. Expected Submission Date									
					Month Day Year				
					N/A N/A N/A				
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)									
On May 12, 2025, at 1353 Central Standard Time (CST), Browns Ferry Nuclear Plant, Unit 2, experienced a significant electro-hydraulic control (EHC) leak, resulting in a turbine trip and automatic reactor SCRAM on low EHC pressure. The trip was not complex with all systems responding normally post trip except for the turbine bypass valves, which subsequently failed closed due to loss of EHC pressure. At 1407 CST, with Unit 2 in Mode 3, there was a second automatic RPS actuation due to a low reactor water level transient caused by manually opening and closing a main steam relief valve. Event Notification 57705 was made to the Nuclear Regulatory Commission (NRC) Operations Center on May 12, 2025, at 1731 CST. It was documented that three broken mounting cap screws were found. Failure analysis was completed on the broken cap screws and determined that the cap screws failed due to hydrogen embrittlement. Investigation into the source of hydrogen intrusion determined that there were two potential sources: environmental intrusion and pickling during the manufacturing process. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of the Reactor Protection System and the Primary Containment Isolation System.									

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: oira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 2	☒ 050	2. DOCKET NUMBER 00260	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Unit 2, was in Mode 1 at 100 percent power.

II. Description of Event**A. Event Summary**

On May 12, 2025, at 1353 Central Standard Time (CST), BFN Unit 2, experienced a significant electro-hydraulic control (EHC)[JA] leak resulting in a turbine trip and automatic reactor SCRAM on low EHC pressure. The trip was not complex with all systems responding normally post trip except for the turbine bypass valves [V], which subsequently failed closed due to loss of EHC pressure. Operations responded and stabilized the plant. At 1407, with Unit 2 in Mode 3, there was a second automatic Reactor Protection System (RPS) actuation due to a low reactor water level transient caused by manually opening and closing a main steam relief valve. Event Notification 57705 was made to the Nuclear Regulatory Commission (NRC) Operations Center.

The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(iv)(A), as any event or condition that resulted in manual or automatic actuation of the Reactor Protection System and the Primary Containment Isolation System (PCIS).

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

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NARRATIVE**C. Dates and approximate times of occurrences**

<u>DATE AND APPROXIMATE TIMES</u>	<u>OCCURRENCE</u>
April 10, 2025, at 2237	Unit 2 mode switch moved to RUN following 2R23 refueling outage
May 12, 2025, at 1353	Turbine Trip on Low EHC Pressure/SCRAM. (Event Notification 57705 was reported to the NRC for this event.)
May 12, 2025, at 1407	RPS actuation due to a low reactor water level transient
May 12, 2025	Walkdown following SCRAM identified two broken mounting cap screws on 2-FCV-47-154C. It was later identified that three cap screws were broken during the event.
May 17, 2025	Following repairs to the valve, the unit was synced to the grid.

D. Manufacturer and model number of each component that failed during the event

Non-safety related bolting for the EHC shutoff valve, QA0 component.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

BFN, Unit 2, experienced a significant EHC leak resulting in a turbine trip and automatic reactor SCRAM on low EHC pressure. Walkdown following SCRAM identified two broken mounting cap screws on 2-FCV-47-154C. It was later identified that three cap screws were broken during the event.

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NARRATIVE**G. The failure mode, mechanism, and effect of each failed component**

The mounting cap screws for BFN-2-047-0154C ultimately failed due to hydrogen embrittlement. Hydrogen embrittlement could have been introduced from environmental sources (plant environment, improper cleaning, improper lubricant, etc.), or during the manufacturing process (black oxide coating).

H. Operator actions

Operations responded and stabilized the plant following the reactor scram and subsequently initiated a plant cooldown to Mode 4.

I. Automatically and manually initiated safety system responses

The trip was not complex with all systems responding normally post trip except for the turbine bypass valves [V], which subsequently failed closed due to loss of EHC pressure. Operations responded and stabilized the plant. At 1407, with Unit 2 in Mode 3, there was second automatic RPS actuation due to a low reactor water level transient caused by manually opening and closing a main steam relief valve. PCIS Groups 2, 3, 6, and 8 isolation signals were received. Upon receipt of these signals, components actuated as required.

III. Cause of the event**A. Cause of each component or system failure or personnel error**

The Unit 2 SCRAM on low EHC pressure was caused by the failure of three of four ASTM A574 grade 4137 black oxide coated mounting cap screws for shutoff valve BFN-2-FCV-047-0154C. Procedure MCI-0-047-VLV002 does not incorporate vendor manual VTD-G080-3255 guidance potentially leading to environmental degradation of the mounting cap screws and introducing another source of hydrogen embrittlement. Hydrogen was potentially introduced during the manufacturing process of coating the mounting cap screws with black oxide.

B. Cause(s) and circumstances for each human performance related root cause

Review of the Human Performance (HU) factors of the event determined that neither the EHC leak nor the associated broken cap screws were the result of a human performance error.

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NARRATIVE**IV. Analysis of the event**

At 13:52 CST, approximately one minute prior to the SCRAM, BFN Operations personnel placed the alternate EHC pump (B) into service for testing. Initial EHC system pressure was approximately 1564 pounds per square inch gauge (PSIG) and increased approximately 54 PSIG (~3.5%) to 1618 PSIG when the B pump was placed in service. A plot of EHC pressure versus time indicates that the leak began when the B pump was placed in service. The EHC control system attempted to recover system pressure via the A and B pumps and accumulators but ultimately failed, and a turbine trip was generated. Although this pressure surge was within EHC operating pressure limits, the pressure surge was the most likely initiating event for the cap screw failures and subsequent SCRAM. Post event walkdowns identified three sheared mounting cap screws on 2-FCV-047-0154C, breaching the EHC pressure boundary and draining the Unit 2 EHC tank.

Operations responded and stabilized the plant. At 1407 CST with Unit 2 in Mode 3, there was a second automatic RPS actuation due to a low reactor water level transient caused by manually opening and closing a main steam relief valve. Reactor water level was maintained via feed water pump. Decay heat was removed by discharging steam (via main steam line drains) to the main condenser. Units 1 and 3 were not affected.

V. Assessment of Safety Consequences

The plant responded as designed, while maintaining defense-in-depth for nuclear safety. All nuclear safety systems functioned as designed. This event was of very low nuclear safety significance. At no time was the health and safety of the public at risk.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

N/A



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NARRATIVE

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

This event did not render a train of a safety system inoperable.

VI. Corrective Actions

Corrective Actions are being managed by the TVA's Corrective Action Program (CAP) under Condition Report (CR) 2012739.

A. Immediate Corrective Actions

Shutoff valve and mounting cap screws were replaced.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

The source of the hydrogen could not be conclusively determined; therefore, the cause evaluation addressed both the potential for environmental sources and manufacturing sources. Potential manufacturing sources will be corrected via Corrective Action to Prevent Recurrence (CAPR) 2012739-005, which will change procurement strategy of control valve, safety valve, bypass valve, and combined intermediate valve mounting cap screws for peripheral valves associated with procedures MCI-0-047-VLV002 and MCI-0-047-VLV001 to QA3 and specify additional documentation and testing requirements. Potential environmental sources will be corrected by CAPR 2012739-007, which will revise procedures MCI-0-047-VLV002 and MCI-0-047-VLV001 to align with vendor manual VTD-G080-3255 recommendations with regards to cleaning, reusing mounting cap screws, using lock washers, and lubricating practices.



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NARRATIVE

VII. Previous Similar Events at the Same Site

There were no previous similar events found at the site.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].